

Audited Corrected Analytical Derivation

Nearest-neighbour ring walk with one absorbing site and one shortcut

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1 Purpose and final conclusions

This document is a detailed audit and correction of the analytical calculation for a nearest-neighbour discrete-time random walk on a ring with one absorbing site and one directed shortcut into the absorbing site. The main conclusion is that the absorbing-ring part of the calculation can be made correct after several index, sign, and orthogonality corrections, but the shortcut-to-target part must be changed more substantially.

The key point is the following. If a shortcut from u to the absorbing target v replaces a fraction β of the stay-put probability $1 - q$, then in the transient state space the shortcut is an extra killing channel at u . Writing

$$\lambda = \beta(1 - q), \quad 0 \leq \beta \leq 1, \quad (1)$$

the corrected resolvent is

$$\boxed{\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) - \frac{z\lambda \tilde{W}_{n_0}(u, z) \tilde{W}_u(n, z)}{1 + z\lambda \tilde{W}_u(u, z)}}. \quad (2)$$

The minus sign and the plus sign in the denominator are both forced by the stochastic interpretation: the shortcut moves probability mass from the transient site u to the absorbing site v .

Consequently, the first-passage generating function is

$$\boxed{\tilde{F}_{n_0}(v, z) = \tilde{F}_{n_0}^{(0)}(v, z) + \frac{z\lambda \tilde{W}_{n_0}(u, z) [1 - \tilde{F}_u^{(0)}(v, z)]}{1 + z\lambda \tilde{W}_u(u, z)}}. \quad (3)$$

Here the superscript (0) means “without the shortcut but with the absorbing target present.” This formula also gives a useful monotonicity check: the shortcut contribution to $\tilde{F}_{n_0}(v, z)$ has non-negative coefficients for $z \in [0, 1)$, so adding a shortcut to the target cannot decrease the probability of having hit the target by any fixed time.

In the symmetric case

$$N = 2L, \quad v = 0, \quad u = L, \quad (4)$$

and for an initial site at distance ρ from v along either half-ring, $0 \leq \rho \leq L$, the corrected first-passage generating function reduces to the closed form

$$\boxed{\tilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}, \quad a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right)}. \quad (5)$$

We use the convention $U_{-1} = 0$. Formula (5) is the cleanest corrected analytical answer.

The poles are the roots of

$$\boxed{aT_L(y_s) + U_{L-1}(y_s) = 0, \quad y_s = 1 + \frac{s-1}{q}.} \quad (6)$$

Equivalently, if $y_s = \cos \theta$, the roots satisfy

$$\boxed{\tan(L\theta) = -a \sin \theta.} \quad (7)$$

The largest positive root s_1 lies in

$$\boxed{\gamma_1 < s_1 < \alpha_1,} \quad \gamma_1 = 1 - q + q \cos \frac{2\pi}{N}, \quad \alpha_1 = 1 - q + q \cos \frac{\pi}{N}. \quad (8)$$

Thus the shortcut-to-target makes the long-time tail decay faster than the no-shortcut absorbing-ring tail. The original condition

$$\beta \leq \frac{2q}{(1-q)N} \quad (9)$$

is not a valid corrected condition for bimodality or pole dominance. The corrected long-time decay is

$$\boxed{F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad t \rightarrow \infty,} \quad (10)$$

not $t\alpha_1^t$. For weak shortcuts,

$$\boxed{s_1 = \alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2),} \quad (11)$$

and therefore

$$\boxed{1 - s_1 = 1 - \alpha_1 + \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2).} \quad (12)$$

For large N this becomes

$$\boxed{1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}.} \quad (13)$$

Equation (13) is the most direct corrected decay-dependence statement for the right tail of the second peak, if a second peak is present.

2 Model and notation

The ring sites are $0, 1, \dots, N-1$, with all site labels understood modulo N . The unmodified nearest-neighbour transition rule is

$$X_{t+1} = \begin{cases} X_t, & \text{with probability } 1-q, \\ X_t + 1, & \text{with probability } q/2, \\ X_t - 1, & \text{with probability } q/2, \end{cases} \quad 0 < q < 1. \quad (14)$$

The absorbing target is denoted by m or, in the shortcut section, by v . If m is absorbing, then once the walker reaches m it is removed from the transient dynamics. The killed propagator is

$$W_{n_0}(n, t) = \Pr\{X_t = n, X_0 = n_0, X_s \neq m \text{ for all } 0 \leq s \leq t\}, \quad (15)$$

for transient sites $n, n_0 \neq m$. The first-passage probability to m is

$$\mathcal{F}_{n_0}(m, t) = \Pr\{X_t = m, X_s \neq m \text{ for } 0 \leq s < t\}, \quad t \geq 1, \quad (16)$$

with $\mathcal{F}_{n_0}(m, 0) = 0$ when $n_0 \neq m$.

Generating functions are defined by

$$\widetilde{W}_{n_0}(n, z) = \sum_{t=0}^{\infty} W_{n_0}(n, t) z^t, \quad \widetilde{F}_{n_0}(m, z) = \sum_{t=0}^{\infty} \mathcal{F}_{n_0}(m, t) z^t. \quad (17)$$

The frequently used change of variable is

$$y = y(z) = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right). \quad (18)$$

It is useful because

$$1 - z[1 - q + q \cos \theta] = zq[y - \cos \theta]. \quad (19)$$

When $N = 2L$ is even, the odd-sector eigenvalues are

$$\alpha_\ell = 1 - q + q \cos \frac{(2\ell - 1)\pi}{N}, \quad \ell = 1, \dots, L, \quad (20)$$

and the even-sector ring eigenvalues are

$$\gamma_r = 1 - q + q \cos \frac{2\pi r}{N}, \quad r = 1, \dots, N - 1. \quad (21)$$

3 Finite-sum identities with details

Let

$$\theta_j = \frac{\pi j}{N}, \quad c_j = \cos \theta_j, \quad A_j(d) = [1 - (-1)^j] \sin(d\theta_j) \sin \theta_j, \quad (22)$$

where $0 < d < N$ is an integer. The central identity is

$$\frac{1}{N} \sum_{j=1}^{N-1} \frac{A_j(d)}{y - c_j} = \frac{U_{N-d-1}(y) + U_{d-1}(y)}{U_{N-1}(y)}. \quad (23)$$

Here T_n and U_n are Chebyshev polynomials of the first and second kind, with

$$T_n(\cos \theta) = \cos(n\theta), \quad U_n(\cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}, \quad U_{-1}(y) = 0. \quad (24)$$

3.1 Proof of the central identity

The denominator $U_{N-1}(y)$ has the simple roots

$$y = c_j = \cos \frac{\pi j}{N}, \quad j = 1, \dots, N - 1. \quad (25)$$

Indeed,

$$U_{N-1}(c_j) = \frac{\sin(N\theta_j)}{\sin \theta_j} = \frac{\sin(\pi j)}{\sin \theta_j} = 0. \quad (26)$$

We compute the residue of the right-hand side of (23) at $y = c_j$. The numerator is

$$\begin{aligned} U_{N-d-1}(c_j) + U_{d-1}(c_j) &= \frac{\sin((N-d)\theta_j) + \sin(d\theta_j)}{\sin \theta_j} \\ &= \frac{\sin(N\theta_j - d\theta_j) + \sin(d\theta_j)}{\sin \theta_j}. \end{aligned} \quad (27)$$

Since $N\theta_j = \pi j$,

$$\begin{aligned} \sin(N\theta_j - d\theta_j) &= \sin(\pi j) \cos(d\theta_j) - \cos(\pi j) \sin(d\theta_j) \\ &= 0 - (-1)^j \sin(d\theta_j) = -(-1)^j \sin(d\theta_j). \end{aligned} \quad (28)$$

Substituting this into (27) gives

$$U_{N-d-1}(c_j) + U_{d-1}(c_j) = \frac{[1 - (-1)^j] \sin(d\theta_j)}{\sin \theta_j}. \quad (29)$$

Next, differentiate $U_{N-1}(\cos \theta) = \sin(N\theta)/\sin \theta$ with respect to y . Because $y = \cos \theta$, $dy/d\theta = -\sin \theta$. Thus

$$\frac{d}{dy} U_{N-1}(\cos \theta) = -\frac{1}{\sin \theta} \frac{d}{d\theta} \left[\frac{\sin(N\theta)}{\sin \theta} \right]. \quad (30)$$

At $\theta = \theta_j$, $\sin(N\theta_j) = 0$ and $\cos(N\theta_j) = (-1)^j$, so

$$\begin{aligned} U'_{N-1}(c_j) &= -\frac{1}{\sin \theta_j} \left[\frac{N \cos(N\theta_j) \sin \theta_j - \sin(N\theta_j) \cos \theta_j}{\sin^2 \theta_j} \right] \\ &= -\frac{1}{\sin \theta_j} \left[\frac{N(-1)^j \sin \theta_j}{\sin^2 \theta_j} \right] = \frac{N(-1)^{j+1}}{\sin^2 \theta_j}. \end{aligned} \quad (31)$$

Therefore the residue of the right-hand side at $y = c_j$ is

$$\begin{aligned} \text{Res}_{y=c_j} \frac{U_{N-d-1}(y) + U_{d-1}(y)}{U_{N-1}(y)} &= \frac{U_{N-d-1}(c_j) + U_{d-1}(c_j)}{U'_{N-1}(c_j)} \\ &= \frac{[1 - (-1)^j] \sin(d\theta_j)}{\sin \theta_j} \frac{\sin^2 \theta_j}{N(-1)^{j+1}}. \end{aligned} \quad (32)$$

If j is even, the factor $1 - (-1)^j$ is zero. If j is odd, then $(-1)^{j+1} = 1$. Hence in all cases

$$\text{Res}_{y=c_j} \frac{U_{N-d-1}(y) + U_{d-1}(y)}{U_{N-1}(y)} = \frac{1}{N} [1 - (-1)^j] \sin(d\theta_j) \sin \theta_j = \frac{A_j(d)}{N}. \quad (33)$$

The left-hand side of (23) has exactly the same simple pole at $y = c_j$ and exactly the same residue $A_j(d)/N$. The difference between the two sides has no finite poles. Both sides tend to zero as $y \rightarrow \infty$. Therefore the difference is an entire rational function tending to zero at infinity, hence it is identically zero. This proves (23).

3.2 The $y \rightarrow 1$ identity

Set $y = 1$ in (23). Since $U_n(1) = n + 1$,

$$U_{N-d-1}(1) + U_{d-1}(1) = (N-d) + d = N, \quad U_{N-1}(1) = N. \quad (34)$$

Thus

$$\boxed{\frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d\theta_j) \sin \theta_j}{1 - \cos \theta_j} = 1.} \quad (35)$$

3.3 The $y = \cos(2\pi r/N)$ identity

For $r = 0, 1, \dots, N-1$, set

$$y = \cos \frac{2\pi r}{N}. \quad (36)$$

If this value is not a pole, the substitution is direct. If it is a zero of U_{N-1} , the value is the removable limit already determined by the residue calculation above. Using the trigonometric form gives

$$\boxed{\frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d\theta_j) \sin \theta_j}{\cos(2\pi r/N) - \cos \theta_j} = \cos \frac{2\pi r d}{N}}. \quad (37)$$

The case $r = 0$ is the same as (35) because $\cos(0) = 1$ and $\cos(0 \cdot d) = 1$.

3.4 The reciprocal Fourier identity

For $j = 1, \dots, N-1$ and $0 \leq d \leq N$, the second identity needed in the absorbing-propagator calculation is

$$\boxed{\frac{1}{N} \sum_{r=1}^{N-1} \frac{\cos(2\pi r d/N)}{\cos \theta_j - \cos(2\pi r/N)} = \frac{1}{N(1 - \cos \theta_j)} - \frac{\sin(d\theta_j)}{\sin \theta_j}}. \quad (38)$$

To see why this is the correct form, include temporarily the omitted $r = 0$ term. Since

$$\frac{1}{N} \frac{1}{\cos \theta_j - 1} = -\frac{1}{N(1 - \cos \theta_j)}, \quad (39)$$

(38) is equivalent to

$$\boxed{\frac{1}{N} \sum_{r=0}^{N-1} \frac{\cos(2\pi r d/N)}{\cos \theta_j - \cos(2\pi r/N)} = -\frac{\sin(d\theta_j)}{\sin \theta_j}}. \quad (40)$$

The right-hand side satisfies the homogeneous second-order recurrence

$$g_{d+1} - 2 \cos \theta_j g_d + g_{d-1} = 0, \quad g_d = -\frac{\sin(d\theta_j)}{\sin \theta_j}, \quad (41)$$

for $1 \leq d \leq N-1$. The Fourier sum on the left-hand side satisfies the same recurrence away from $d = 0$ because multiplication by $2 \cos(2\pi r/N)$ shifts the Fourier modes. The boundary condition is also consistent because $N\theta_j = \pi j$ implies

$$g_{N-d} = -\frac{\sin(N\theta_j - d\theta_j)}{\sin \theta_j} = -\frac{-(-1)^j \sin(d\theta_j)}{\sin \theta_j} = (-1)^j \frac{\sin(d\theta_j)}{\sin \theta_j}. \quad (42)$$

In the only use below, this identity is multiplied by $1 - (-1)^j$, so only odd j contributes, and for odd j the symmetry becomes $g_{N-d} = g_d$, as required by the cosine Fourier sum. This establishes (38) in the sector used by the calculation. Equivalently, one may derive it from the partial-fraction expansion of $[T_N(x) - 1]^{-1}$.

4 Absorbing ring without shortcut

4.1 Probability-conserving ring propagator

The probability-conserving ring propagator is

$$Q_{n_0}(n, t) = \frac{1}{N} \sum_{r=0}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} \gamma_r^t, \quad (43)$$

where $\gamma_0 = 1$ and γ_r for $r \geq 1$ is given by (21). Separating the zero mode gives

$$Q_{n_0}(n, t) = \frac{1}{N} + \frac{1}{N} \sum_{r=1}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} \gamma_r^t. \quad (44)$$

4.2 First-passage probability to a single absorbing site

Let the absorbing site be m and define

$$d_0 = |n_0 - m|, \quad 0 < d_0 < N. \quad (45)$$

The first-passage probability to m on the ring without shortcut is

$$\boxed{\mathcal{F}_{n_0}(m, t) = \frac{q}{N} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \lambda_j^{t-1}, \quad t \geq 1,} \quad (46)$$

where

$$\lambda_j = 1 - q + q \cos \theta_j. \quad (47)$$

If $N = 2L$, only the odd j values contribute, and (46) becomes

$$\mathcal{F}_{n_0}(m, t) = \sum_{\ell=1}^L f_\ell(n_0, m) \alpha_\ell^{t-1}, \quad (48)$$

with

$$\boxed{f_\ell(n_0, m) = \frac{2q}{N} \sin \frac{(2\ell - 1)\pi |n_0 - m|}{N} \sin \frac{(2\ell - 1)\pi}{N}.} \quad (49)$$

The generating function follows directly from (19):

$$\begin{aligned} \tilde{\mathcal{F}}_{n_0}(m, z) &= z \frac{q}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - z \lambda_j} \\ &= \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{y - \cos \theta_j} \\ &= \boxed{\frac{U_{N-d_0-1}(y) + U_{d_0-1}(y)}{U_{N-1}(y)}}. \end{aligned} \quad (50)$$

Setting $z = 1$ gives $y = 1$, and (35) gives

$$\tilde{\mathcal{F}}_{n_0}(m, 1) = 1. \quad (51)$$

Thus the absorbing target is eventually reached with probability one.

4.3 Survival probability

The survival probability up to and including time t is

$$R_{n_0}(t) = 1 - \sum_{s=0}^t \mathcal{F}_{n_0}(m, s). \quad (52)$$

Using (46), $\mathcal{F}_{n_0}(m, 0) = 0$, and the finite geometric sum

$$\sum_{s=1}^t \lambda_j^{s-1} = \frac{1 - \lambda_j^t}{1 - \lambda_j} = \frac{1 - \lambda_j^t}{q(1 - \cos \theta_j)}, \quad (53)$$

we get

$$\begin{aligned} \sum_{s=1}^t \mathcal{F}_{n_0}(m, s) &= \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} (1 - \lambda_j^t) \\ &= 1 - \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} \lambda_j^t, \end{aligned} \quad (54)$$

where the last step used (35). Hence

$$R_{n_0}(t) = \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} \lambda_j^t. \quad (55)$$

4.4 Renewal equation and the killed propagator

For $t \geq 1$, the killed propagator satisfies the renewal relation

$$W_{n_0}(n, t) = Q_{n_0}(n, t) - \sum_{s=1}^t \mathcal{F}_{n_0}(m, s) Q_m(n, t - s). \quad (56)$$

This subtracts all paths that first hit m at time s and then, in the unrestricted ring, continue from m to n in the remaining $t - s$ steps.

Insert (44) and (46). The zero Fourier mode of Q_m gives

$$\frac{1}{N} \sum_{s=1}^t \mathcal{F}_{n_0}(m, s) = \frac{1}{N} [1 - R_{n_0}(t)]. \quad (57)$$

This cancels the $1/N$ term in $Q_{n_0}(n, t)$ and leaves $R_{n_0}(t)/N$, which is precisely the second line in (59). For the nonzero Fourier modes, we need

$$\begin{aligned} \sum_{s=1}^t \lambda_j^{s-1} \gamma_r^{t-s} &= \gamma_r^{t-1} + \lambda_j \gamma_r^{t-2} + \dots + \lambda_j^{t-1} \\ &= \frac{\gamma_r^t - \lambda_j^t}{\gamma_r - \lambda_j} \\ &= \frac{\gamma_r^t - \lambda_j^t}{q[\cos(2\pi r/N) - \cos \theta_j]}. \end{aligned} \quad (58)$$

Because this convolution is subtracted in (56), the denominator can be rewritten as

$$\cos \theta_j - \cos(2\pi r/N),$$

giving the fully expanded expression

$$\begin{aligned} W_{n_0}(n, t) = & \frac{1}{N} \sum_{r=1}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} \gamma_r^t \\ & + \frac{1}{N^2} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} \lambda_j^t \\ & + \frac{1}{N^2} \sum_{j=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \cos \frac{2\pi r(n-m)}{N}}{\cos \theta_j - \cos(2\pi r/N)} (\gamma_r^t - \lambda_j^t). \end{aligned} \quad (59)$$

The sign of the last bracket is important. It must be $\gamma_r^t - \lambda_j^t$ when the denominator is $\cos \theta_j - \cos(2\pi r/N)$.

Now reduce the double sum. The part proportional to γ_r^t uses (37) with $d = d_0$:

$$\begin{aligned} & \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{\cos \theta_j - \cos(2\pi r/N)} \\ & = -N \cos \frac{2\pi r d_0}{N}. \end{aligned} \quad (60)$$

Therefore the γ_r^t part of the double sum is

$$-N \sum_{r=1}^{N-1} \cos \frac{2\pi r d_0}{N} \cos \frac{2\pi r(n-m)}{N} \gamma_r^t. \quad (61)$$

The part proportional to $-\lambda_j^t$ uses (38) with $d = |n - m|$:

$$\begin{aligned} & - \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \lambda_j^t \sum_{r=1}^{N-1} \frac{\cos(2\pi r(n-m)/N)}{\cos \theta_j - \cos(2\pi r/N)} \\ & = - \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \lambda_j^t \left[\frac{1}{1 - \cos \theta_j} - N \frac{\sin(d \theta_j)}{\sin \theta_j} \right] \\ & = - \sum_{j=1}^{N-1} [1 - (-1)^j] \left[\frac{\sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} - N \sin(d_0 \theta_j) \sin(d \theta_j) \right] \lambda_j^t. \end{aligned} \quad (62)$$

After substituting (61) and (62) into (59), the survival term cancels the first term inside the square brackets of (62). The final killed propagator is

$$\boxed{W_{n_0}(n, t) = \frac{1}{N} \sum_{r=1}^{N-1} \left[\cos \frac{2\pi r(n - n_0)}{N} - \cos \frac{2\pi r(n - m)}{N} \cos \frac{2\pi r(n_0 - m)}{N} \right] \gamma_r^t + \frac{1}{N} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin \frac{\pi j |n_0 - m|}{N} \sin \frac{\pi j |n - m|}{N} \lambda_j^t.} \quad (63)$$

4.5 Boundary and initial-condition checks

If $n = m$, then the sine factor in the second line of (63) is zero and the first-line bracket becomes

$$\cos \frac{2\pi r(m - n_0)}{N} - 1 \cdot \cos \frac{2\pi r(n_0 - m)}{N} = 0. \quad (64)$$

So $W_{n_0}(m, t) = 0$, as required for a killed propagator. Similarly, if $n_0 = m$, both lines vanish for transient n .

At $t = 0$, for transient $n, n_0 \neq m$, (63) gives the identity $W_{n_0}(n, 0) = \delta_{n, n_0}$. Here are the details. Let

$$d = |n - m|, \quad d_0 = |n_0 - m|. \quad (65)$$

The basic Fourier identity is

$$\frac{1}{N} \sum_{r=1}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} = \delta_{n, n_0} - \frac{1}{N}. \quad (66)$$

The product-to-sum formula gives

$$\begin{aligned} \sum_{r=1}^{N-1} \cos \frac{2\pi r d}{N} \cos \frac{2\pi r d_0}{N} &= \frac{1}{2} \sum_{r=1}^{N-1} \left[\cos \frac{2\pi r(d - d_0)}{N} + \cos \frac{2\pi r(d + d_0)}{N} \right] \\ &= \frac{N}{2} (\delta_{d, d_0} + \delta_{d, N-d_0}) - 1, \end{aligned} \quad (67)$$

where the Kronecker deltas are interpreted modulo the ring. The odd sine sum is

$$\begin{aligned} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d\theta_j) \sin(d_0\theta_j) &= 2 \sum_{\substack{1 \leq j \leq N-1 \\ j \text{ odd}}} \sin(d\theta_j) \sin(d_0\theta_j) \\ &= \frac{N}{2} (\delta_{d, d_0} + \delta_{d, N-d_0}). \end{aligned} \quad (68)$$

Substituting (67) and (68) into (63) at $t = 0$ cancels the distance-only terms and leaves δ_{n, n_0} .

4.6 Generating-function form

Define

$$G_r^{(1)}(n_0, n, m) = \frac{1}{N} \left[\cos \frac{2\pi r(n - n_0)}{N} - \cos \frac{2\pi r(n - m)}{N} \cos \frac{2\pi r(n_0 - m)}{N} \right], \quad (69)$$

$$G_j^{(2)}(n_0, n, m) = \frac{1}{N} [1 - (-1)^j] \sin \frac{\pi j |n_0 - m|}{N} \sin \frac{\pi j |n - m|}{N}. \quad (70)$$

Then

$$W_{n_0}(n, t) = \sum_{r=1}^{N-1} G_r^{(1)}(n_0, n, m) \gamma_r^t + \sum_{j=1}^{N-1} G_j^{(2)}(n_0, n, m) \lambda_j^t. \quad (71)$$

If $N = 2L$, the second sum can be written over odd modes only:

$$W_{n_0}(n, t) = \sum_{r=1}^{N-1} G_r^{(1)}(n_0, n, m) \gamma_r^t + \sum_{\ell=1}^L g_\ell^{(2)}(n_0, n, m) \alpha_\ell^t, \quad (72)$$

where

$$g_\ell^{(2)}(n_0, n, m) = \frac{2}{N} \sin \frac{(2\ell-1)\pi |n_0 - m|}{N} \sin \frac{(2\ell-1)\pi |n - m|}{N}. \quad (73)$$

The generating function is therefore

$$\widetilde{W}_{n_0}(n, z) = \sum_{r=1}^{N-1} \frac{G_r^{(1)}(n_0, n, m)}{1 - z\gamma_r} + \sum_{\ell=1}^L \frac{g_\ell^{(2)}(n_0, n, m)}{1 - z\alpha_\ell}. \quad (74)$$

4.7 Special symmetric return propagator

Let $N = 2L$, $m = 0$, and $n = n_0 = L$. Then the first line of (63) vanishes because

$$\cos \frac{2\pi r(L-L)}{N} = 1, \quad \cos \frac{2\pi rL}{N} \cos \frac{2\pi rL}{N} = \cos^2(\pi r) = 1. \quad (75)$$

The second line becomes

$$W_L(L, t) = \frac{1}{N} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin^2 \frac{\pi j}{2} \lambda_j^t. \quad (76)$$

Only odd $j = 2\ell - 1$ contribute. For odd j , $1 - (-1)^j = 2$ and $\sin^2(\pi j/2) = 1$, so

$$W_L(L, t) = \frac{2}{N} \sum_{\ell=1}^L \alpha_\ell^t. \quad (77)$$

Hence

$$\widetilde{W}_L(L, z) = \frac{2}{N} \sum_{\ell=1}^L \frac{1}{1 - z\alpha_\ell} = \frac{2}{N} \frac{1}{zq} \sum_{\ell=1}^L \frac{1}{y - \cos \frac{(2\ell-1)\pi}{2L}}. \quad (78)$$

The zeros of $T_L(y)$ are exactly $\cos[(2\ell-1)\pi/(2L)]$, so

$$\frac{T'_L(y)}{T_L(y)} = \sum_{\ell=1}^L \frac{1}{y - \cos \frac{(2\ell-1)\pi}{2L}}. \quad (79)$$

Since $T'_L(y) = LU_{L-1}(y)$ and $N = 2L$, (78) gives

$$\widetilde{W}_L(L, z) = \frac{2}{2L} \frac{1}{zq} \frac{LU_{L-1}(y)}{T_L(y)} = \frac{U_{L-1}(y)}{zq T_L(y)}. \quad (80)$$

Using $U_{2L-1}(y) = 2T_L(y)U_{L-1}(y)$, this is also

$$\boxed{\widetilde{W}_L(L, z) = \frac{2U_{L-1}^2(y)}{zq U_{2L-1}(y)}}. \quad (81)$$

At $z = 1$, $y = 1$, so

$$\widetilde{W}_L(L, 1) = \frac{U_{L-1}(1)}{qT_L(1)} = \frac{L}{q} = \frac{N}{2q}. \quad (82)$$

5 Shortcut into the absorbing site

5.1 Transient matrix derivation of the sign

Let K_0 be the transient transition matrix for the ring with absorbing target v , but without the shortcut. Its resolvent is

$$R_0(z) = (I - zK_0)^{-1}, \quad [R_0(z)]_{n,n_0} = \widetilde{W}_{n_0}(n, z). \quad (83)$$

A shortcut from u to the absorbing target v that replaces probability $\lambda = \beta(1 - q)$ of the stay-put move at u removes that probability from the transient state u . Therefore, in the transient subspace,

$$K = K_0 - \lambda|u\rangle\langle u|. \quad (84)$$

The resolvent with the shortcut is

$$R(z) = (I - zK)^{-1} = [I - zK_0 + z\lambda|u\rangle\langle u|]^{-1}. \quad (85)$$

Set

$$A = I - zK_0. \quad (86)$$

Then

$$R(z) = [A + z\lambda|u\rangle\langle u|]^{-1}. \quad (87)$$

Use the Sherman–Morrison identity

$$(A + |p\rangle\langle q|)^{-1} = A^{-1} - \frac{A^{-1}|p\rangle\langle q|A^{-1}}{1 + \langle q|A^{-1}|p\rangle}. \quad (88)$$

Here $|p\rangle = z\lambda|u\rangle$ and $\langle q| = \langle u|$. Thus

$$R(z) = R_0(z) - \frac{z\lambda R_0(z)|u\rangle\langle u|R_0(z)}{1 + z\lambda\langle u|R_0(z)|u\rangle}. \quad (89)$$

Taking the (n, n_0) matrix element gives the corrected propagator

$$\boxed{\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\lambda \widetilde{W}_{n_0}(u, z) \widetilde{W}_u(n, z)}{1 + z\lambda \widetilde{W}_u(u, z)}}. \quad (90)$$

This is the same as (2). The sign is not a convention: it is fixed by the fact that the target state v is absorbing and therefore absent from the transient subspace.

5.2 First-passage generating function

Sum (90) over all transient sites $n \neq v$. For the baseline killed walk,

$$\sum_{n \neq v} \widetilde{W}_r(n, z) = \frac{1 - \widetilde{F}_r^{(0)}(v, z)}{1 - z}. \quad (91)$$

Therefore

$$\sum_{n \neq v} \widetilde{S}_{n_0}(n, z) = \frac{1 - \widetilde{F}_{n_0}^{(0)}(v, z)}{1 - z} - \frac{z\lambda \widetilde{W}_{n_0}(u, z)}{1 + z\lambda \widetilde{W}_u(u, z)} \frac{1 - \widetilde{F}_u^{(0)}(v, z)}{1 - z}. \quad (92)$$

But the left-hand side is also $(1 - \widetilde{F}_{n_0}(v, z))/(1 - z)$. Multiplying by $1 - z$ and rearranging gives

$$\boxed{\widetilde{F}_{n_0}(v, z) = \widetilde{F}_{n_0}^{(0)}(v, z) + \frac{z\lambda \widetilde{W}_{n_0}(u, z) [1 - \widetilde{F}_u^{(0)}(v, z)]}{1 + z\lambda \widetilde{W}_u(u, z)}}. \quad (93)$$

5.3 Why the sign is also forced physically

For $0 \leq z < 1$, all coefficients of $\widetilde{W}_{n_0}(u, z)$ and $1 - \widetilde{F}_u^{(0)}(v, z)$ are non-negative, and $1 + z\lambda\widetilde{W}_u(u, z) > 0$. Hence the shortcut term in (93) is non-negative. Thus adding a shortcut into the absorbing site increases the first-passage generating function, as it should.

By contrast, a formula with

$$+ \frac{z\lambda\widetilde{W}_{n_0}(u, z)\widetilde{W}_u(n, z)}{1 - z\lambda\widetilde{W}_u(u, z)} \quad (94)$$

in the propagator would increase transient survival and decrease the hitting generating function. It can also produce a positive real singularity when

$$z\lambda\widetilde{W}_u(u, z) = 1. \quad (95)$$

In the symmetric case, $\widetilde{W}_L(L, 1) = N/(2q)$, so the denominator at $z = 1$ would be

$$1 - \frac{\beta(1-q)N}{2q}, \quad (96)$$

which is exactly the source of the original threshold. Such a singularity is an artefact of the wrong sign; it is not present in the corrected stochastic shortcut-to-target model.

6 Symmetric shortcut: exact corrected first-passage generating function

We now set

$$N = 2L, \quad v = 0, \quad u = L. \quad (97)$$

Let $F_\rho(z)$ be the first-passage generating function to 0 from a site at distance ρ from 0 along either half-ring, where $0 \leq \rho \leq L$. By symmetry, $F_{L+k} = F_{L-k}$.

6.1 Difference equation

For $1 \leq \rho \leq L-1$, the first step gives

$$F_\rho(z) = z \left[(1-q)F_\rho(z) + \frac{q}{2}F_{\rho-1}(z) + \frac{q}{2}F_{\rho+1}(z) \right]. \quad (98)$$

Move all terms to one side and divide by $zq/2$:

$$F_{\rho+1}(z) - 2yF_\rho(z) + F_{\rho-1}(z) = 0, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right). \quad (99)$$

The absorbing boundary is

$$F_0(z) = 1. \quad (100)$$

At $\rho = L$, the shortcut replaces probability $\lambda = \beta(1-q)$ of the stay-put probability by an immediate jump to the target. Therefore

$$F_L(z) = z [\lambda + (1-q-\lambda)F_L(z) + qF_{L-1}(z)]. \quad (101)$$

Using $1-q-\lambda = (1-\beta)(1-q)$ and writing

$$h = \frac{\lambda}{q} = \frac{\beta(1-q)}{q}, \quad a = \frac{1}{h} = \frac{q}{\beta(1-q)}, \quad (102)$$

the boundary equation becomes

$$\boxed{(y+h)F_L(z) - F_{L-1}(z) = h.} \quad (103)$$

6.2 Solving the recurrence

The recurrence (99) is the Chebyshev recurrence. The solution satisfying $F_0 = 1$ can be written as

$$F_\rho = T_\rho(y) + C U_{\rho-1}(y), \quad (104)$$

because $T_0 = 1$ and $U_{-1} = 0$. Hence

$$F_L = T_L + C U_{L-1}, \quad F_{L-1} = T_{L-1} + C U_{L-2}. \quad (105)$$

Substitute these into (103):

$$\begin{aligned} h &= (y + h)(T_L + C U_{L-1}) - (T_{L-1} + C U_{L-2}) \\ &= (y T_L - T_{L-1}) + C(y U_{L-1} - U_{L-2}) + h T_L + h C U_{L-1}. \end{aligned} \quad (106)$$

The needed Chebyshev identities are

$$y U_{L-1} - U_{L-2} = T_L, \quad y T_L - T_{L-1} = (y^2 - 1) U_{L-1}. \quad (107)$$

Using them in (106) gives

$$h = (y^2 - 1) U_{L-1} + C T_L + h T_L + h C U_{L-1}. \quad (108)$$

Therefore

$$C(T_L + h U_{L-1}) = h(1 - T_L) - (y^2 - 1) U_{L-1}, \quad (109)$$

and

$$C = \frac{h(1 - T_L) - (y^2 - 1) U_{L-1}}{T_L + h U_{L-1}}. \quad (110)$$

Substitution into (104) gives one correct form of the answer:

$$F_\rho = T_\rho + U_{\rho-1} \frac{h(1 - T_L) - (y^2 - 1) U_{L-1}}{T_L + h U_{L-1}}. \quad (111)$$

Multiplying numerator and denominator by $a = 1/h$ gives

$$F_\rho = \frac{a T_\rho T_L + T_\rho U_{L-1} + U_{\rho-1}(1 - T_L) - a(y^2 - 1) U_{\rho-1} U_{L-1}}{a T_L + U_{L-1}}. \quad (112)$$

Now use the two product identities

$$T_\rho T_L - (y^2 - 1) U_{\rho-1} U_{L-1} = T_{L-\rho}, \quad (113)$$

$$T_\rho U_{L-1} + U_{\rho-1}(1 - T_L) = U_{L-\rho-1} + U_{\rho-1}. \quad (114)$$

For example, if $y = \cos \theta$, then (113) reads

$$\cos(\rho\theta) \cos(L\theta) + \sin(\rho\theta) \sin(L\theta) = \cos((L - \rho)\theta), \quad (115)$$

and (114) follows from

$$\cos(\rho\theta) \frac{\sin(L\theta)}{\sin \theta} + \frac{\sin(\rho\theta)}{\sin \theta} [1 - \cos(L\theta)] = \frac{\sin((L - \rho)\theta) + \sin(\rho\theta)}{\sin \theta}. \quad (116)$$

Thus (112) becomes

$$\boxed{\tilde{F}_\rho(z) = \frac{a T_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{a T_L(y) + U_{L-1}(y)}}. \quad (117)$$

This is the same as (5).

6.3 Checks on the symmetric formula

First, $\rho = 0$ gives

$$\tilde{F}_0(z) = \frac{aT_L + U_{-1} + U_{L-1}}{aT_L + U_{L-1}} = 1, \quad (118)$$

as required for a walker already at the absorbing site. Second, in the no-shortcut limit $\beta \rightarrow 0$, we have $a \rightarrow \infty$, so

$$\tilde{F}_\rho(z) \rightarrow \frac{T_{L-\rho}(y)}{T_L(y)}. \quad (119)$$

This agrees with the absorbing-ring formula (50). Indeed, for $N = 2L$,

$$\frac{U_{2L-\rho-1} + U_{\rho-1}}{U_{2L-1}} = \frac{T_{L-\rho}}{T_L}, \quad (120)$$

which follows from the same trigonometric product identities.

Third, when $\rho = L$,

$$\tilde{F}_L(z) = \frac{a + U_{L-1}(y)}{aT_L(y) + U_{L-1}(y)}. \quad (121)$$

At $z = 1$, $y = 1$, so $T_L(1) = 1$ and $U_{L-1}(1) = L$, giving

$$\tilde{F}_L(1) = \frac{a + L}{a + L} = 1. \quad (122)$$

Thus absorption remains certain.

7 Poles, tail decay, and the corrected dependence on β

7.1 Pole equation

Write $s = 1/z$, so

$$y_s = 1 + \frac{s - 1}{q}. \quad (123)$$

The poles of (117) are roots of

$$D(y_s) = aT_L(y_s) + U_{L-1}(y_s) = 0. \quad (124)$$

If $y_s = \cos \theta$, then

$$T_L(y_s) = \cos(L\theta), \quad U_{L-1}(y_s) = \frac{\sin(L\theta)}{\sin \theta}. \quad (125)$$

Therefore (124) becomes

$$a \cos(L\theta) + \frac{\sin(L\theta)}{\sin \theta} = 0. \quad (126)$$

Multiplying by $\sin \theta$ and dividing by $\cos(L\theta)$ away from the asymptotes gives

$$\boxed{\tan(L\theta) = -a \sin \theta.} \quad (127)$$

The corresponding pole value is

$$\boxed{s = 1 - q + q \cos \theta.} \quad (128)$$

7.2 Location of the largest pole

The no-shortcut largest odd-sector eigenvalue is

$$\alpha_1 = 1 - q + q \cos \frac{\pi}{2L} = 1 - q + q \cos \frac{\pi}{N}. \quad (129)$$

The largest even-sector nontrivial eigenvalue is

$$\gamma_1 = 1 - q + q \cos \frac{\pi}{L} = 1 - q + q \cos \frac{2\pi}{N}. \quad (130)$$

Consider the interval

$$\frac{\pi}{2L} < \theta < \frac{\pi}{L}. \quad (131)$$

In this interval,

$$\cos(L\theta) < 0, \quad \sin(L\theta) > 0, \quad \sin \theta > 0. \quad (132)$$

Hence $D(\cos \theta) = a \cos(L\theta) + \sin(L\theta)/\sin \theta$ tends to $U_{L-1} > 0$ at the left endpoint $\theta = \pi/(2L)$, while it equals $a \cos \pi = -a < 0$ at the right endpoint $\theta = \pi/L$. By continuity there is a root θ_1 in this interval. This root gives

$$\cos \frac{\pi}{L} < \cos \theta_1 < \cos \frac{\pi}{2L}, \quad (133)$$

and therefore

$$\boxed{\gamma_1 < s_1 < \alpha_1.} \quad (134)$$

Since the transient transition matrix is non-negative and primitive for $0 < q < 1$, its spectral radius is the unique positive dominant eigenvalue. Thus s_1 controls the long-time tail.

7.3 Partial fractions and the exact tail coefficient

Let

$$N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y), \quad D(y) = aT_L(y) + U_{L-1}(y). \quad (135)$$

Let y_k be a simple root of $D(y)$ and define

$$s_k = 1 - q + qy_k. \quad (136)$$

Near $s = s_k$,

$$\tilde{F}_\rho(1/s) = \frac{N_\rho(y_s)}{D(y_s)} = \frac{N_\rho(y_k)}{D'(y_k)(y_s - y_k)} + \text{regular terms}. \quad (137)$$

Since $y_s - y_k = (s - s_k)/q$, the residue in the s -plane is

$$\text{Res}_{s=s_k} \tilde{F}_\rho(1/s) = \frac{qN_\rho(y_k)}{D'(y_k)}. \quad (138)$$

The partial-fraction term corresponding to a pole s_k is

$$\frac{B_{\rho k}}{1 - zs_k}, \quad (139)$$

whose $s = 1/z$ form is

$$\frac{B_{\rho k}}{1 - s_k/s} = \frac{B_{\rho k}s}{s - s_k}. \quad (140)$$

Its residue at $s = s_k$ is $B_{\rho k} s_k$. Hence the coefficient multiplying s_k^t is

$$B_{\rho k}^{(z^t)} = \frac{q N_\rho(y_k)}{s_k D'(y_k)}. \quad (141)$$

Equivalently, if we write the time dependence as s_k^{t-1} for $t \geq 1$, then

$$F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \quad B_{\rho k} = \frac{q N_\rho(y_k)}{D'(y_k)}. \quad (142)$$

In particular,

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad B_{\rho 1} = \frac{q N_\rho(y_1)}{D'(y_1)}. \quad (143)$$

The decay time can be reported as

$$\tau_\beta = -\frac{1}{\log s_1}. \quad (144)$$

7.4 Weak-shortcut expansion

For small β , equivalently large

$$a = \frac{q}{\beta(1-q)}, \quad (145)$$

the largest root lies close to

$$\theta_0 = \frac{\pi}{2L}. \quad (146)$$

Write

$$\theta_1 = \theta_0 + \delta, \quad 0 < \delta \ll 1. \quad (147)$$

Then

$$L\theta_1 = \frac{\pi}{2} + L\delta, \quad \tan(L\theta_1) = \tan\left(\frac{\pi}{2} + L\delta\right) = -\cot(L\delta). \quad (148)$$

For small δ ,

$$-\cot(L\delta) = -\frac{1}{L\delta} + O(L\delta). \quad (149)$$

The root equation (127) gives

$$-\frac{1}{L\delta} + O(\delta) = -a \sin(\theta_0 + \delta). \quad (150)$$

To leading order,

$$\delta = \frac{1}{aL \sin \theta_0} + O(a^{-2}). \quad (151)$$

Now expand $s_1 = 1 - q + q \cos(\theta_0 + \delta)$:

$$\begin{aligned} s_1 &= 1 - q + q [\cos \theta_0 - \delta \sin \theta_0 + O(\delta^2)] \\ &= \alpha_1 - q \sin \theta_0 \frac{1}{aL \sin \theta_0} + O(a^{-2}) \\ &= \alpha_1 - \frac{q}{aL} + O(a^{-2}). \end{aligned} \quad (152)$$

Since $q/a = \beta(1 - q)$ and $L = N/2$,

$$s_1 = \alpha_1 - \frac{2\beta(1 - q)}{N} + O([\beta(1 - q)]^2). \quad (153)$$

Thus

$$1 - s_1 = 1 - \alpha_1 + \frac{2\beta(1 - q)}{N} + O([\beta(1 - q)]^2). \quad (154)$$

For large N ,

$$1 - \alpha_1 = q \left[1 - \cos \frac{\pi}{N} \right] = \frac{q\pi^2}{2N^2} + O(N^{-4}), \quad (155)$$

so

$$1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1 - q)}{N}. \quad (156)$$

7.5 Strong-shortcut endpoint expansion

When a is small, the largest root approaches the zero of U_{L-1} at $\theta = \pi/L$. Write

$$\theta_1 = \frac{\pi}{L} - \varepsilon, \quad 0 < \varepsilon \ll 1. \quad (157)$$

Then

$$\tan(L\theta_1) = \tan(\pi - L\varepsilon) = -L\varepsilon + O(\varepsilon^3), \quad (158)$$

and

$$-a \sin \theta_1 = -a \sin \frac{\pi}{L} + O(a\varepsilon). \quad (159)$$

Thus

$$\varepsilon = \frac{a}{L} \sin \frac{\pi}{L} + O(a^2). \quad (160)$$

Consequently

$$\begin{aligned} s_1 &= 1 - q + q \cos \left(\frac{\pi}{L} - \varepsilon \right) \\ &= \gamma_1 + q\varepsilon \sin \frac{\pi}{L} + O(\varepsilon^2) \\ &= \gamma_1 + \frac{qa}{L} \sin^2 \frac{\pi}{L} + O(a^2). \end{aligned} \quad (161)$$

This shows explicitly how the dominant pole moves from α_1 toward γ_1 as the shortcut strength increases.

8 Consequences for the bimodality discussion

The existence of a second peak is a finite-time shape question. It is not determined solely by which pole is asymptotically largest. In the corrected formula, if

$$F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \quad (162)$$

then local extrema are controlled exactly by the finite difference

$$F_\rho(t+1) - F_\rho(t) = \sum_k B_{\rho k} (s_k - 1) s_k^{t-1}. \quad (163)$$

A second peak at time t_* requires

$$F_\rho(t_*) - F_\rho(t_* - 1) > 0, \quad F_\rho(t_* + 1) - F_\rho(t_*) < 0. \quad (164)$$

These inequalities can be checked exactly from (163). They do not reduce to the original threshold $\beta \leq 2q/[(1-q)N]$.

What can be stated analytically and cleanly is the decay of the right tail of a second peak if it is present:

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad \frac{F_\rho(t+1)}{F_\rho(t)} \rightarrow s_1. \quad (165)$$

Thus the referee-facing corrected decay dependence is

$$\tau_\beta^{-1} = -\log s_1, \quad aT_L \left(1 + \frac{s_1 - 1}{q}\right) + U_{L-1} \left(1 + \frac{s_1 - 1}{q}\right) = 0. \quad (166)$$

For weak shortcuts this gives

$$\tau_\beta^{-1} = -\log \left[\alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2) \right]. \quad (167)$$

Equivalently, in gap form,

$$1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}. \quad (168)$$

The shortcut contribution to the decay gap is the term

$$\frac{2\beta(1-q)}{N}. \quad (169)$$

9 Audit notes: issues found in the original calculation

This section records the mathematical and typographical issues identified during the audit.

9.1 Substantive mathematical issues

1. The shortcut-to-target resolvent sign is wrong for the stochastic interpretation in which a fraction $\beta(1-q)$ of the stay-put probability at u is replaced by a jump to absorbing v . The corrected transient perturbation is negative killing at u , giving (90).
2. The original denominator $1 - z\beta(1-q)\widetilde{W}_u(u, z)$ must be replaced by $1 + z\beta(1-q)\widetilde{W}_u(u, z)$. This changes the pole equation from a positive horizontal-line intersection to a negative one, or equivalently from $a - \phi = 0$ to $a + \phi = 0$.
3. The original first-passage correction has the wrong monotonicity: it decreases the hitting generating function after adding a shortcut to the target. The corrected formula (93) increases it.
4. The original beta condition

$$\beta \leq \frac{2q}{(1-q)N} \quad (170)$$

comes from the wrong-sign denominator and is not a valid corrected condition for the stochastic shortcut-to-target model.

5. The original $t\alpha_1^t$ conclusion comes from an artificial collision between an uncorrected shortcut pole and an absorbing-ring pole. In the corrected model the dominant pole is simple and equals $s_1 < \alpha_1$, so the tail is $B_{\rho 1} s_1^{t-1}$.
6. The reduction of the absorbing-ring double sum contains a sign inconsistency in the intermediate line: with denominator $\cos(\pi j/N) - \cos(2\pi r/N)$, the bracket must be $\gamma_r^t - \lambda_j^t$. The final killed propagator is nevertheless correct after this sign is fixed.
7. The initial-condition check following the killed propagator uses incorrect individual orthogonality claims. The correct cosine and odd-sine sums are (67) and (68); only their combination gives δ_{n,n_0} .
8. In the long convolution formula corresponding to the original Eq. (41), the first no-shortcut term should involve $f_\ell(n_0, v)$, not $f_\ell(n_0, u)$. The baseline term in $F_{n_0}(v, t)$ is the first-passage probability to v .
9. In the same long convolution formula, terms involving the odd-sector propagator $g_r^{(2)}$ should carry α_r , not γ_r . The displayed use of γ_r in the last line is an index/eigenvalue mismatch.
10. The residue coefficient formula for the uncorrected c_k contains bracket mismatches and an apparently incorrect derivative factor. Since that coefficient belongs to the wrong-sign pole problem, it should be replaced rather than repaired.

9.2 Typographical and presentation issues

1. The document class option `idelinks` should be `hidelinks`.
2. The source depends on local style files such as `customcolours.sty`, `customcommands.sty`, and `globalplotconfig.sty`; a portable derivation should avoid these dependencies or include them.
3. A label of the form `\label{eq:}` is not useful and can create reference problems.
4. Several expressions write $G_k^{(1)}(n_0, n, , m)$ and $G_k^{(2)}(n_0, n, , m)$ with a double comma.
5. The text says $Q_{n_0}(n, 0) = \delta_{n,n_0}$ and $Q_{n_0}(n, t) = 0$ in places where it means $W_{n_0}(n, 0)$ and $W_{n_0}(n, t)$.
6. The compact odd-sector sums require N even. This assumption should be stated before using $\sum_{\ell=1}^{N/2}$.
7. The figure caption says “given the requirement $q > 1$ ”; the stochastic model requires $0 < q < 1$, and a nonzero shortcut probability requires $q < 1$ if the shortcut probability is proportional to $1 - q$.
8. The phrase “poles in the dominator” should be “poles in the denominator.”
9. The contour statement after $z = 1/s$ should specify orientation and which singularities are enclosed. The corrected derivation avoids this ambiguity by using partial fractions directly in s .

10 Optional numerical check for the figure parameters

For the original figure parameters

$$N = 10, \quad L = 5, \quad q = \frac{2}{3}, \quad \beta = \frac{2}{7}, \quad (171)$$

we have

$$a = \frac{q}{\beta(1-q)} = 7. \quad (172)$$

The corrected largest root solves

$$\tan(5\theta) = -7 \sin \theta, \quad \frac{\pi}{10} < \theta < \frac{\pi}{5}. \quad (173)$$

Numerically,

$$\theta_1 \approx 0.3865953134, \quad s_1 = 1 - \frac{2}{3} + \frac{2}{3} \cos \theta_1 \approx 0.9507987460. \quad (174)$$

The no-shortcut largest odd eigenvalue is

$$\alpha_1 = 1 - \frac{2}{3} + \frac{2}{3} \cos \frac{\pi}{10} \approx 0.9673710109, \quad (175)$$

so indeed

$$s_1 < \alpha_1. \quad (176)$$

The corrected tail gap is

$$1 - s_1 \approx 0.0492012540. \quad (177)$$